# THE DERIVED CATEGORY OF MODULES SUPPORTED ON A CLOSED SUBSET IS EQUIVALENT TO THE DERIVED CATEGORY OF THE CLOSED SUBSPACE, WITH THE FUNCTOR AS RIGHT ADJOINT RECONSTRUCTION

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites three cross-chapter labels, each recorded in assembly.external_references with its canonical tag; none replaces a load-bearing step, which are all inside the span.

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*Allowed standard prerequisites (editorial).*

- Exactness of pushforward and pullback along a closed immersion. Stacks Project, Modules on Sites, Lemmas modules-lemma-exactness-pushforward-pullback (tag 01AJ) and modules-lemma-i-star-exact (tag 01AX). Supplies exactness of i_* and i^* = i^{-1}, which drives full faithfulness of i_*.
- Cohomology with support on a sheaf supported in the closed subset. Stacks Project, Cohomology of Sheaves, Lemma lemma-cohomology-with-support-sheaf-on-support (tag 0A3B). Identifies R-H_Z with i^{-1} on D_Z(O_X) and gives part (3).

*Source reference.* Original lemma 0A3B: sources/stacks-chow/tags.txt, source lines 12734–12734. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01AJ: sources/stacks-chow/tags.txt, source lines 1692–1692. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01AX: sources/stacks-chow/tags.txt, source lines 1706–1706. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $i : Z \to X$ be the inclusion of a closed subset.*

(1) *For $K$ in $D(\mathcal{O}_X|_Z)$ we have $i_*K$ in $D_Z(\mathcal{O}_X)$.*
(2) *The functor $i_* : D(\mathcal{O}_X|_Z) \to D_Z(\mathcal{O}_X)$ is an equivalence with quasi-inverse $i^{-1}|_{D_Z(\mathcal{O}_X)} = R\mathcal{H}_Z|_{D_Z(\mathcal{O}_X)}$.*
(3) *The functor $i_* \circ R\mathcal{H}_Z : D(\mathcal{O}_X) \to D_Z(\mathcal{O}_X)$ is right adjoint to the inclusion functor $D_Z(\mathcal{O}_X) \to D(\mathcal{O}_X)$.*

*Proof.* Part (1) is immediate from the definitions. Part (3) is a formal consequence of part (2) and Lemma 0A3B. In the rest of the proof we prove part (2).

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Let us think of $i$ as the morphism of ringed spaces $i : (Z, \mathcal{O}_X|_Z) \to (X, \mathcal{O}_X)$. Recall that $i^*$ and $i_*$ is an adjoint pair of functors. Since $i$ is a closed immersion, $i_*$ is exact. Since $i^{-1}\mathcal{O}_X = \mathcal{O}_X|_Z$ is the structure sheaf of $(Z, \mathcal{O}_X|_Z)$ we see that $i^* = i^{-1}$ is exact and we see that that $i^*i_* = i^{-1}i_*$ is isomorphic to the identify functor. See Modules, Lemmas 01AJ and 01AX. Thus $i_* : D(\mathcal{O}_X|_Z) \to D_Z(\mathcal{O}_X)$ is fully faithful and $i^{-1}$ determines a left inverse. On the other hand, suppose that $K$ is an object of $D_Z(\mathcal{O}_X)$ and consider the adjunction map $K \to i_*i^{-1}K$. Using exactness of $i_*$ and $i^{-1}$ this induces the adjunction maps $H^n(K) \to i_*i^{-1}H^n(K)$ on cohomology sheaves. Since these cohomology sheaves are supported on $Z$ we see these adjunction maps are isomorphisms and we conclude that $i_* : D(\mathcal{O}_X|_Z) \to D_Z(\mathcal{O}_X)$ is an equivalence.

To finish the proof it suffices to show that $R\mathcal{H}_Z(K) = i^{-1}K$ if $K$ is an object of $D_Z(\mathcal{O}_X)$. To do this we can use that $K = i_*i^{-1}K$ as we've just proved this is the case. Then Lemma 0A3B tells us what we want. $\square$